



SUMMARY

A passionate Mechanical and Aerospace postdoctoral researcher with 6 years of academic research experience in the fields of Unmanned Aerial Vehicles, Space Weather Data Assimilation, and Machine Learning techniques

EDUCATION

Degree	Major	University	GPA	Month/Year
Doctoral of Philosophy	Aerospace Engineering	Illinois Institute of Technology	3.74/4.00	May/2024
Master of Engineering	Mechanical Engineering	Illinois Institute of Technology	4.00/4.00	May/2020
Bachelor of Science	Mechanical Engineering	Illinois Institute of Technology	3.84/4.00	May/2020

RESEARCH EXPERIENCE

SUPERScript NUMBERS ARE ASSOCIATED WITH PUBLICATIONS, POSTERS & PRESENTATIONS

Center for Space and Atmospheric Research, Embry-Riddle Aeronautical University Jun 2024 - Current
Postdoctoral Researcher, Advisor: Alan Liu & Wenjun Dong, Funded by NSF & NASA Daytona Beach, FL

- ⁶Contributing to the development of CAM-NET, an AI-driven foundation model for high-precision atmospheric prediction. Trained on datasets from the Whole Atmosphere Community Climate Model with thermosphere and ionosphere extension (WACCM-X), the model achieves a computational speedup of 3–4 orders of magnitude compared to conventional simulations and enables rapid generation of atmospheric state forecasts.
- ⁵Developing a Physics-Informed Multi-Variable Transformer Model (PI-MVTI), a machine learning framework for reconstructing missing atmospheric LiDAR data. The model integrates physical constraints—such as mass conservation and buoyancy—to ensure physical consistency and improve accuracy in variables such as temperature and density.

Space Weather Lab, Illinois Institute of Technology Jun 2021 - Jun 2024
Graduate Research Assistant, Advisor: Seebany Datta-Barua, funded by NASA Chicago, IL

- Developed the EMPIRE algorithm (Estimating Model Parameters from Ionospheric Reverse Engineering), a Kalman filter-based data assimilation method designed to reduce discrepancies between observations and model outputs.
- ⁴Investigated the storm-driven phenomenon known as Nighttime Ionospheric Localized Enhancement (NILE), characterized by increased electron density over Florida during geomagnetic storms. Visualized and analyzed outputs to support proposed mechanisms.
- ³Integrated a Kalman smoother into EMPIRE for bidirectional processing to improve state estimation accuracy.
- ²Designed and evaluated representation error to improve error covariance matrices, validating performance using statistical metrics such as mean error and standard deviation.
- ¹Processed and augmented ion drift measurements from SuperDARN and Millstone Hill; validated results against independent datasets.

High Altitude Observatory, National Center Atmospheric Research Jun 2023 - Aug 2023
Graduate Visitor Program Fellow, Advisor: Nicholas Pedatella, funded by NSF Boulder, CO

- Implemented the WACCM-X model coupled with the ensemble-based Data Assimilation Research Testbed (DART) for upper atmospheric simulation and assimilation.
- ⁴Visualized vector maps of atmospheric drivers using Python (Matplotlib) and analyzed the mechanisms behind NILE.

Robotics Lab, Illinois Institute of Technology Jun 2019 - May 2021
Research Assistant, Advisor: Matthew Spenko Chicago, IL

- Used CNNs (AlexNet, GoogLeNet, VGG-16, ResNet) to classify tree species from UAV imagery, applying data augmentation and hyperparameter tuning to improve validation accuracy from 84% to 92%.
- Designed a drone-mounted robotic arm for tree sampling; tested performance and stability in real-world conditions.
- Integrated a RealSense stereo camera and Nvidia Jetson Xavier for detecting tree lichens via OpenCV; reduced static lichen localization error by 50% using a Kalman filter.

PUBLICATIONS

Journal Papers

Published

- ²Hu, J., López Rubio, A., Chartier, A., McDonald, S., & Datta-Barua, S. (2024). *Quantification of representation error in the neutral winds and ion drifts using data assimilation*. Space Weather, 22, e2023SW003609. <https://doi.org/10.1029/2023SW003609>
- ¹Hu, J., McDonald, S., Chartier, A., López Rubio, A., & Datta-Barua, S. (2024). *Data assimilation of ion drift measurements for estimation of ionospheric plasma drivers*. Space Weather, 22, e2024SW003933. <https://doi.org/10.1029/2024SW003933>

Under review

- ⁵**Hu, J.**, Dong W., Liu A. “A Physics Informed Multi-variable Transformer Model for Imputing Missing Sodium LiDAR Observations over the Andes”

In preparation

- ⁴**Hu J.**, López Rubio A., McDonald S. E., Chartier A., Datta-Barua S. "Investigation of Nighttime Ionospheric Localized Enhancement (NILE) Driving Mechanism via Data Assimilation Algorithms"
- ⁶**Hu J.**, Dong W. “CAM-NET: An AI Foundation Model for Whole Atmosphere with Thermosphere and Ionosphere Extension”
- ³**Hu J.**, Datta-Barua S. “Development of a Filtering-Smoothing Technique for Reducing High-latitude Errors in a Data Assimilation Algorithm”

Conference Papers

Published

- ¹**Hu Jiahui**, Rubio Aurora López, Chartier Alex, Bust Gary S., Datta-Barua Seebany “Data Assimilation of Ion Drift Measurements for Estimation of Ionospheric Plasma Drivers”, *Proceedings of the 34th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2021)*, St. Louis, Missouri, September 2021, pp. 3833-3847. <https://doi.org/10.33012/2021.18031>
- ⁴**Hu Jiahui**, McDonald Sarah E., Rubio Aurora López, Chartier Alex, Bust Gary S., Datta-Barua Seebany "Investigating Possible Causes of the Nighttime Ionospheric Localized Enhancement (NILE) via Data Assimilation," *Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2022)*, Denver, Colorado, September 2022, pp. 2989-3001. <https://doi.org/10.33012/2022.18342>

CONFERENCE & WORKSHOP ACTIVITIES

Posters

- ⁶**J. Hu**, W. Dong, A. Liu (2025) *CAM-NET: An AI foundation model for whole atmosphere predictions*. Coupling, Energetics and Dynamics of Atmospheric Region (CEDAR) program, Des Moines, IA, USA. <https://cedarscience.org/poster-2025/cam-net-ai-foundation-model-whole-atmosphere-predictions>
- ⁶**J. Hu**, W. Dong, A. Liu (2025) *A physics-informed multi-variable transformer model for imputing missing sodium LiDAR observations over the Andes*. Coupling, Energetics and Dynamics of Atmospheric Region (CEDAR) program, Des Moines, IA, USA. <https://cedarscience.org/poster-2025/physics-informed-multi-variable-transformer-model-imputing-missing-sodium-lidar>
- ⁵**J. Hu**, W. Dong, A. Liu (2024) *A Physics-informed Multi-variable Transformer Model for Imputing Missing Sodium LiDAR Observations Over the Andes*. American Geophysical Union (AGU) 2024 Meeting, Washington D.C., USA. <https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1523612>
- ³**J. Hu**, E. McDonald, A. Chartier, S. Datta-Barua (2024) *A synthetic study on Kalman smoother coupling to a data assimilation algorithm*. Coupling, Energetics and Dynamics of Atmospheric Region (CEDAR) program, San Diego, CA, USA. <https://cedarscience.org/sites/default/files/2024-posters/Jiahui-Hu.pdf>
- ⁴**J. Hu**, N. M. Pedatella, S. E. McDonald, A. Chartier, S. Datta-Barua (2023) *Nighttime Ionospheric Localized Enhancement (NILE) Mechanism via WACCM-x + DART and EMPIRE Data Assimilation Algorithms*. American Geophysical Union (AGU) 2023 Meeting, San Francisco, CA, USA. <https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1374629>
- ³**J. Hu**, A. Lopez Rubio, S. Datta-Barua (2023) *Filtering/Smoothing Coupled Technique Implementation to the data assimilation algorithm*. Coupling, Energetics and Dynamics of Atmospheric Region (CEDAR) program, San Diego, CA, USA. <https://cedarscience.org/sites/default/files/2023-posters/DATA-15-Hu-Jiahui-DA-Kalman-Smoother.pdf>
- ¹**J. Hu**, A. Lopez Rubio, S. Datta-Barua (2022) *Storm-time Ion Drifts Estimation via a New Augmented Data Assimilation Algorithm*. Coupling, Energetics and Dynamics of Atmospheric Region (CEDAR) program, Austin, TX, USA. https://cedarscience.org/sites/default/files/2022-posters/ITIT-11_Hu_Wednesday.pdf

Presentations

- ⁶**J. Hu**, W. Dong, A. Liu (2025) *CAM-NET: An AI foundation model for whole atmosphere predictions*. CAM-NET: An AI foundation model for whole atmosphere predictions (CESM) 2025 workshop, Boulder, CO, USA.
- ^{1,3}**J. Hu**, S. Datta-Barua (2024) *Improving a data assimilation algorithm EMPIRE via data augmentation and Kalman smoother integration*. Geospace Data Assimilation Working Group (GeoDAWG), virtual. **Invited talk.**
- ¹**J. Hu**, S. Datta-Barua (2024) *Developments in Data Assimilation Algorithms for Ion Drift Estimations Using a New Augmentation Method*. Stanford Center for Position, Navigation and Time (SCPNT). Palo Alto, CA, USA. **Invited talk.**
- ¹**J. Hu**, S. Datta-Barua (2024) *Data assimilation of ion drift measurements for estimation of ionospheric plasma drivers*. Coupling, Energetics and Dynamics of Atmospheric Region Program (CEDAR), San Diego, CA, USA. **Invited talk.**
- ²**J. Hu**, A. Lopez Rubio, S. Datta-Barua (2023) *Observation Error Quantification and Analysis in Data Assimilation*. Advancing National Space Weather Expertise and Research toward Societal Resilience (ANSWERS) student workshop. Virtual. **Invited talk.**
- ⁴**J. Hu**, S. McDonald, A. Lopez Rubio, A. Chartier, Bust G. S. , S. Datta-Barua, (2022) *Investigating Possible Causes of the Nighttime Ionospheric Localized Enhancement (NILE) via Data Assimilation*. The 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2022), Virtual.

TRAINING EXPERIENCE

Incoherent Scatter Radar Summer School, Boston University

Jul 17- Jul 22 2022

Attendee

Boston, MA

- Gained knowledge from lectures covering topics such as incoherent scatter radar (ISR) theory, and IDR operations and analysis techniques.
- Engaged in collaborating closely with ISR facility experts in areas of proposal design and data analysis.
- Acquired practical experience from conducting experiments with fellow attendees and delivering joint findings to the entire group.

HAARP / SAGO Research Seminar, University of Alaska Fairbanks

Oct 11 - Oct 15 2021

Attendee

Fairbanks & Gakona, AK

- Participated in a seminar offering exposure to Subauroral Geophysical Observatory (SAGO) and the High-frequency Active Auroral Research Program (HAARP).
- Recognized the facility's capacity to explore nonlinear effects in plasma, setting it apart from other research facilities.
- Understood that the HAARP facility's unique phased array of antennas allows for beam formation at multiple frequencies and rapid sky scanning.

TEACHING EXPERIENCE

Mechanical Material Aerospace Engineering Department, Illinois Institute of Technology

Aug 2020 - May 2021

Teaching Assistant, Course: Design for Innovation

Chicago, IL

- Provided support to approximately 50 students per term in the completion of hands-on projects, such as foam chairs, trebuchets, and walking robots, as part of the Mechanical Design for Innovation course. This assistance encompassed guiding students through the processes of developing workflows, creating engineering drawings, and coding controllers.

CERTIFICATIONS & AWARDS

- Awarded National Center Atmospheric Research Graduate Visitor Program (NCAR GVP) fellowship with the acceptance rate of 13/87 (June 2023)
- Bachelor's degree graduated Magna Cum Laude (Spring 2020)
- Phi Theta Kappa Member: Distinguished Chapter Award (2019), Honors in Action Project Award (2017)
- Undergraduate Dean's list (Fall 2017, Spring 2018, Spring 2019, Fall 2019)

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, C++, Fortran
- **Tools and Frameworks:** Creo/Inventor, ANSYS, Jupyter, GitHub, Visual Studio
- **Operating Systems:** Windows, MacOS, Linux

REFERENCES LIST

Name	Professional Title	Organization	Work Email
Seebany Datta-Barua	Professor of Mechanical and Aerospace Engineering	Illinois Institute of Technology	sdattaba@iit.edu
Nicholas Pedatella	Scientist II at High Altitude Observatory	National Center Atmospheric Research	nickp@ucar.edu
Alex Chartier	Staff Scientist in Ionospheric Physics at Applied Physics Laboratory	Johns Hopkins University	Alex.Chartier@jhuapl.edu